

Ravi Singh

Senior Software Engineer · Tech Lead - Fintech & Payments Platforms

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PROFESSIONAL SUMMARY

Senior Java Backend Engineer and Tech Lead with **9+ years** in fintech, core banking, and PCI-DSS payments security. At Osfin, independently architected a tokenization platform processing **50M+ card transactions** under PCI-DSS Level 1, re-engineered the core reconciliation engine to cut **10M-record (1 crore)** runtime from **30 minutes to 4 minutes - a 7.5x speedup**, and delivered cryptographic key rotation for **10M cards in under 1 minute** across OCI, AWS, GCP, and Azure. Tech Owner for 2-3 banking clients with full production accountability - monitoring, deployments, migrations, and P1 incident response. Active user of AI-assisted engineering (Claude, GitHub Copilot).

CORE COMPETENCIES

PCI-DSS L1 Tokenization

Fintech Reconciliation

Multi-Cloud (OCI / AWS / GCP / Azure)

Kubernetes & Microservices

Distributed Messaging (Kafka / RabbitMQ)

Cryptographic Key Management

Tech Lead & Multi-Bank Ownership

AI-Assisted Engineering

WORK EXPERIENCE

Osfin

Senior Software Engineer (Tech Lead) · Bengaluru, India

April 2023 - Present

Osfin is an intelligent financial reconciliation and operations automation platform serving banking, fintech, and payments clients globally via 170+ data connectors. Leading a 9-engineer cross-functional team (5 backend, 2 frontend, 2 QA); Tech Owner for 2-3 banking clients.

- Architected** an end-to-end PCI-DSS L1 tokenization engine processing **50M+ card transactions** at sub-millisecond resolution - replaced raw PANs with secure tokens across all payment touchpoints.
- Re-architected** the core reconciliation engine from batch DB queries to in-memory processing - reduced **10M-record (1 crore)** runtime from **30 minutes to 4 minutes (7.5x speedup)**.
- Delivered** cryptographic key rotation for **10M cards in under 1 minute** via a high-concurrency rotation pipeline with automated secret injection and zero customer impact.
- Engineered** cross-cloud key management spanning **OCI KMS, AWS KMS, GCP Cloud KMS, and Azure Key Vault** - zero-downtime key lifecycle operations across all four major clouds.
- Built** asynchronous event ingestion across **Apache Kafka, RabbitMQ, OCI Queue, and Azure Service Bus** - millions of multi-bank events daily with ordered delivery and DLQ recovery.
- Achieved** 99.5%+ auto-match accuracy and reduced manual reconciliation effort by **70%** via event-driven automation; integrated HashiCorp Vault for dynamic secrets and encryption-as-a-service.
- Owned production operations** for 2-3 banking clients as Tech Owner: 24/7 monitoring, schema and infra migrations with **<5-minute downtime windows**, RCA on P1 incidents, **zero critical vulnerabilities** at go-live and through audits.
- Containerized** services with Docker; orchestrated zero-downtime blue-green deployments on OKE / EKS / GKE via Jenkins and GitHub Actions; mentored 5 engineers on PCI-DSS, secure coding, and microservices design.

Java Full Stack Developer · Pune, India

Bee-Recon (client: Games24x7) - enterprise banking reconciliation for foreign-currency accounts, direct payments, cash, card terminals (Visa, Mastercard, Eurocard), and bank transfers. Core Banking Management System (client: Sanjivani Credit Cooperative Society).

- **Engineered** high-volume reconciliation engine processing **300K+ daily transactions** across 8+ payment channels with sub-second match latency.
- **Improved** batch processing throughput by **45%** by redesigning Spring Batch pipelines with parallel step execution and chunk-based processing.
- **Developed** RESTful microservices integrating 5+ third-party banking APIs; reduced integration failures by **60%** via retry logic and dead-letter queues.
- **Implemented** Spring Security RBAC and Log4j2 structured logging - cut incident resolution time by **35%**.
- **Built** a full-stack core banking platform (Spring + Hibernate backend with Angular 4 frontend) covering account management, deposits, loans, and transaction workflows for a cooperative bank.

EDUCATION

Bachelor of Engineering - Electronics and Communication · Apex Institute of Engineering & Technology (AIET), Jaipur

2017

TECHNICAL SKILLS

Languages: Java 8/11/17, SQL, TypeScript, JavaScript

Backend & Frameworks: Spring Boot, Spring Security, Spring Batch, Spring IoC/AOP, Hibernate, REST, Microservices, Event-Driven Architecture

Cloud: OCI (OKE, KMS, Queue, Secrets) · AWS (EKS, EC2, S3, SQS, Secrets Manager, SSM) · GCP (GKE, Cloud KMS) · Azure (AKS, Key Vault, Service Bus)

Messaging & Streaming: Apache Kafka, RabbitMQ, OCI Queue, Azure Service Bus, AWS SQS

Security & Compliance: PCI-DSS Level 1, Tokenization, AES-256, RSA, HSM-backed Key Management, HashiCorp Vault, Multi-Cloud KMS, Zero-Trust

Databases: Oracle DB, MySQL, SQL Server

DevOps & Observability: Docker, Kubernetes (OKE / EKS / GKE / AKS), Jenkins, GitHub Actions, Maven, Git, Prometheus, Grafana, ELK

AI Tools: Claude (Anthropic), GitHub Copilot

Leadership: Tech Lead (9 engineers), Tech Owner (2-3 banking clients), Architecture Reviews, Threat Modeling, Production Incident Response, Stakeholder Management